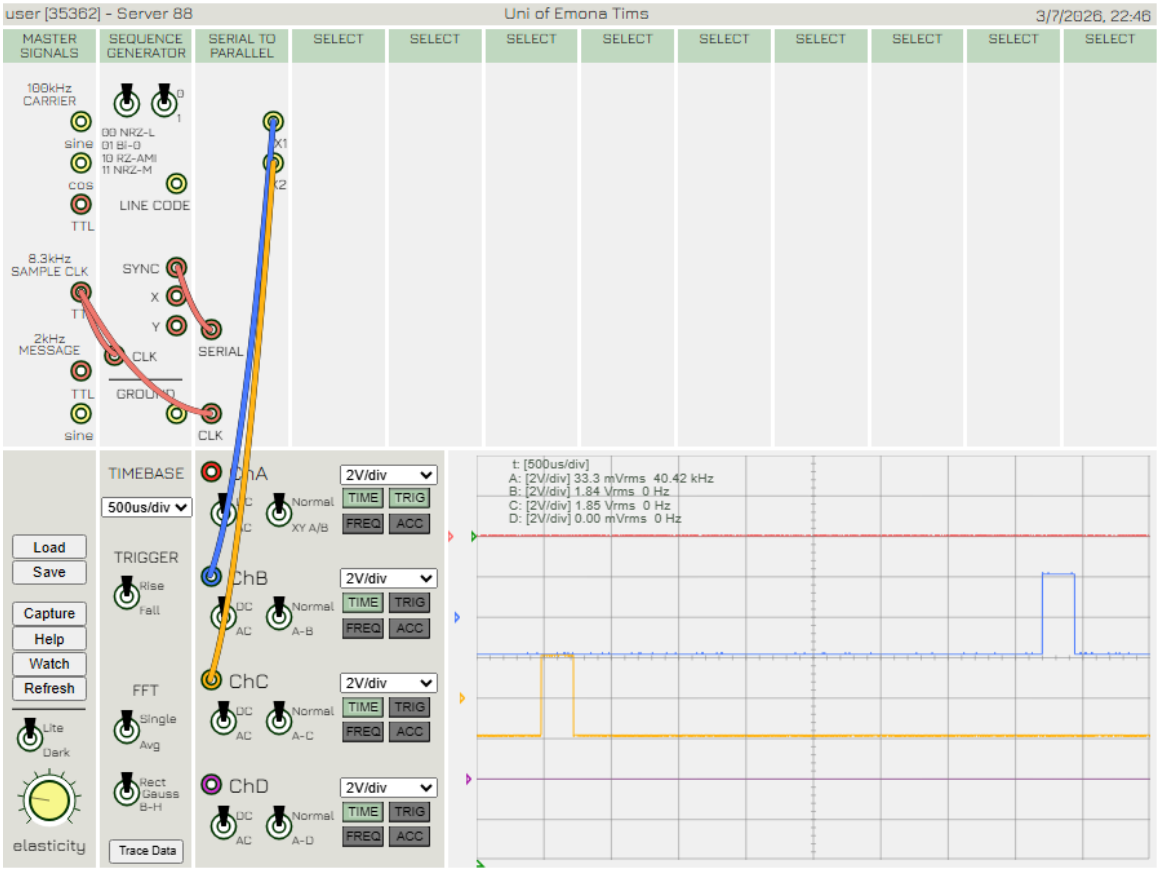
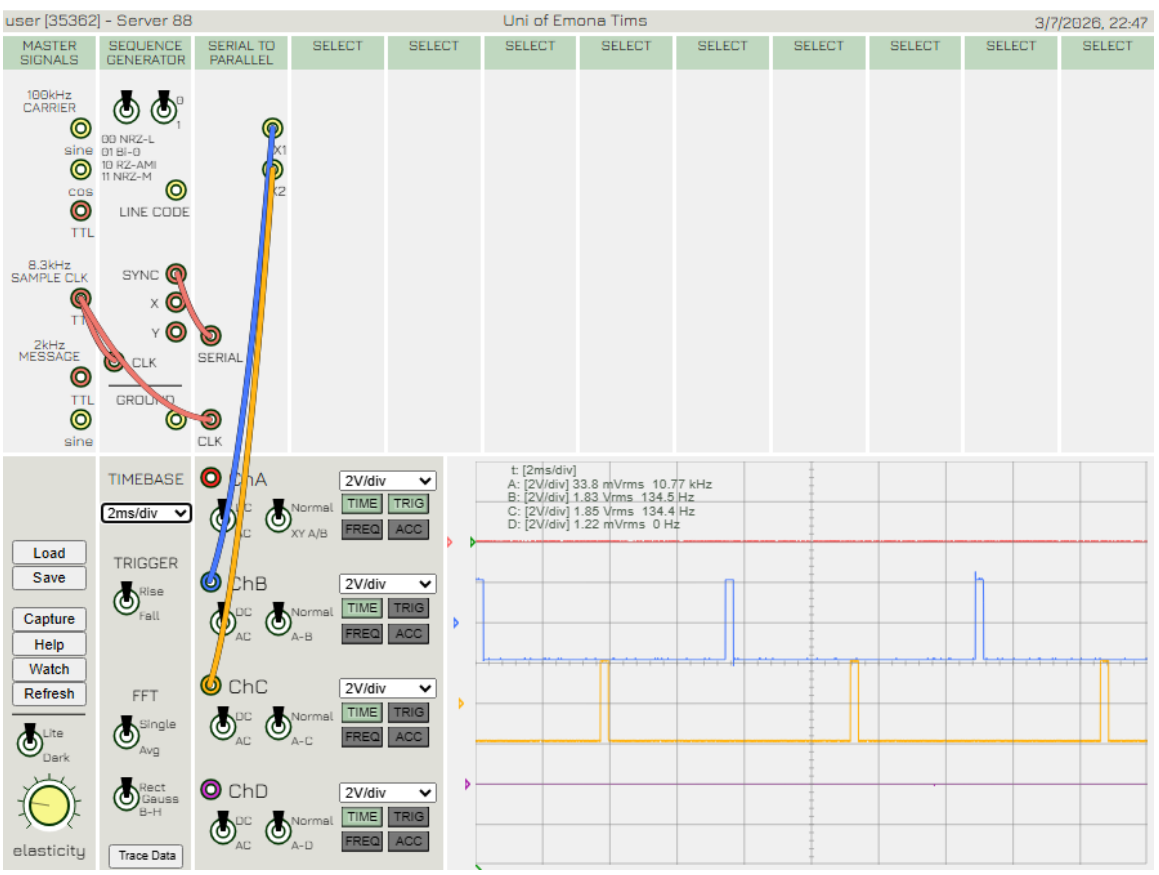
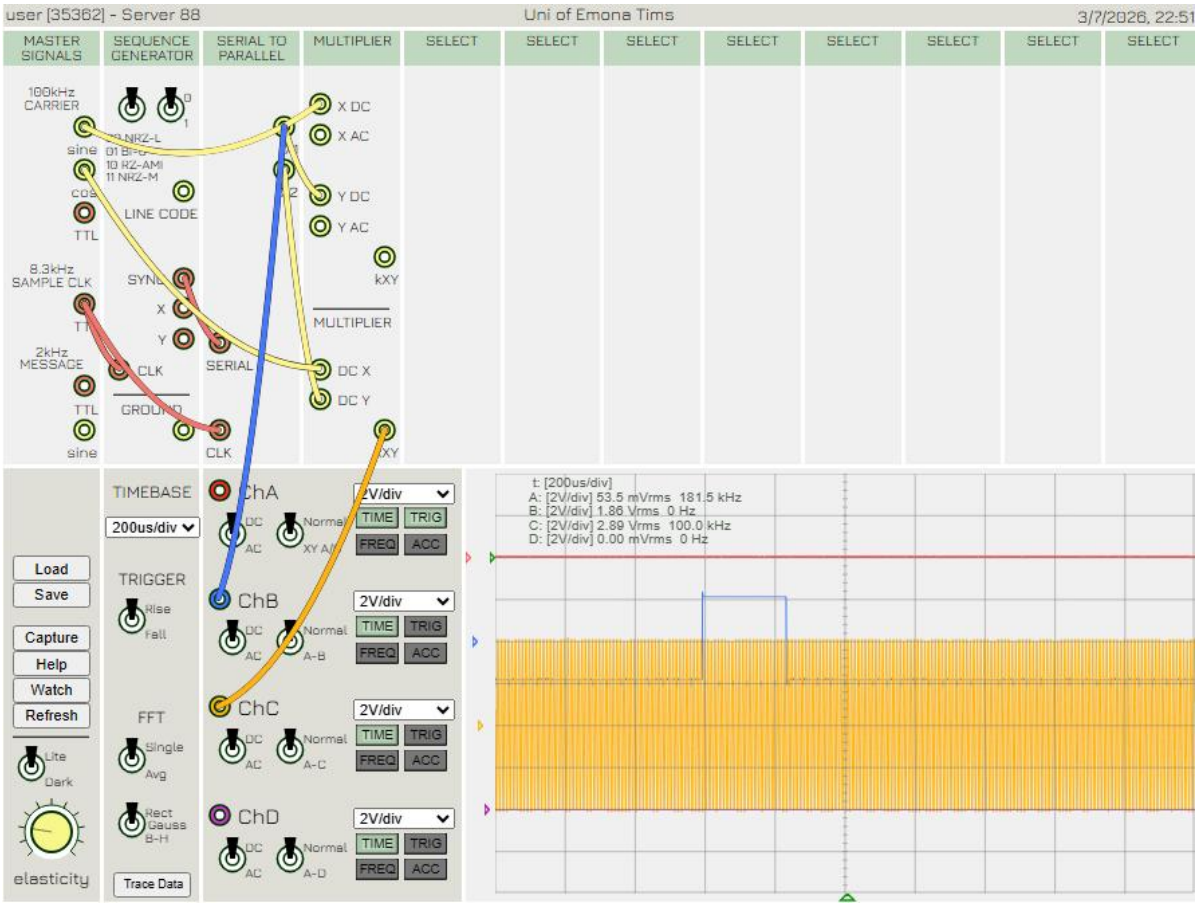
Experiment 18 – A (Generating a QPSK signal)

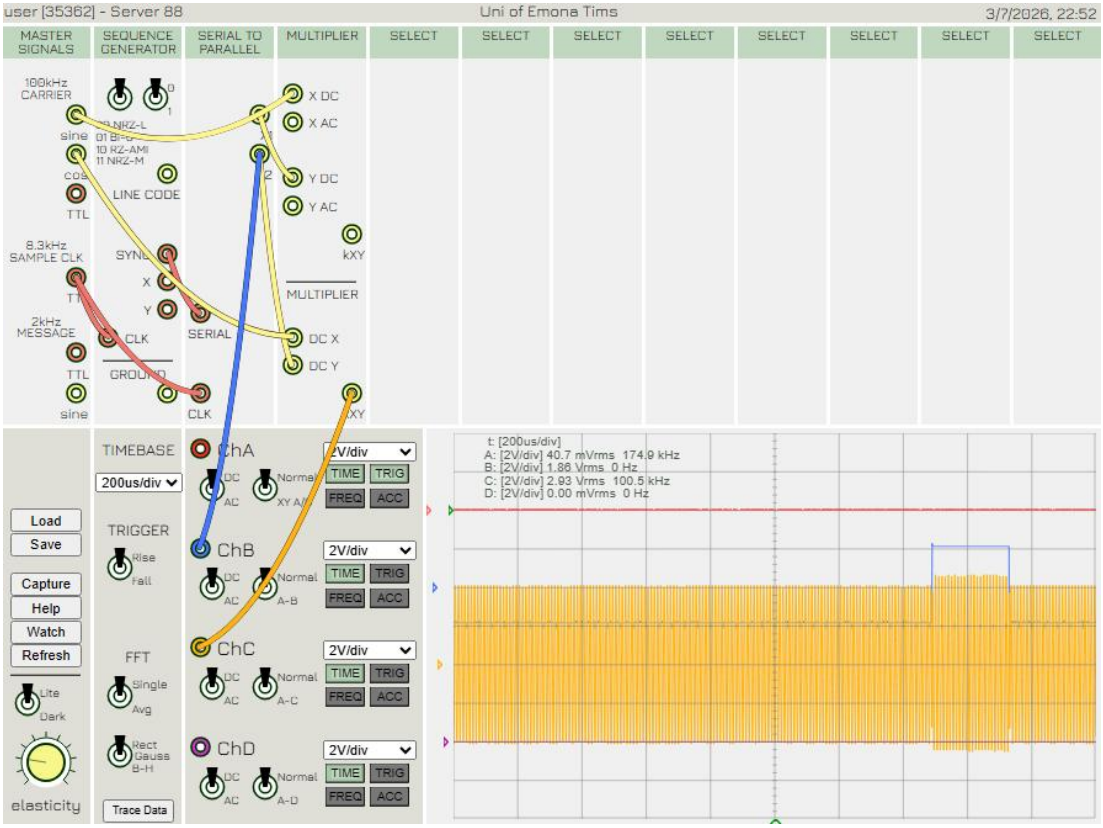




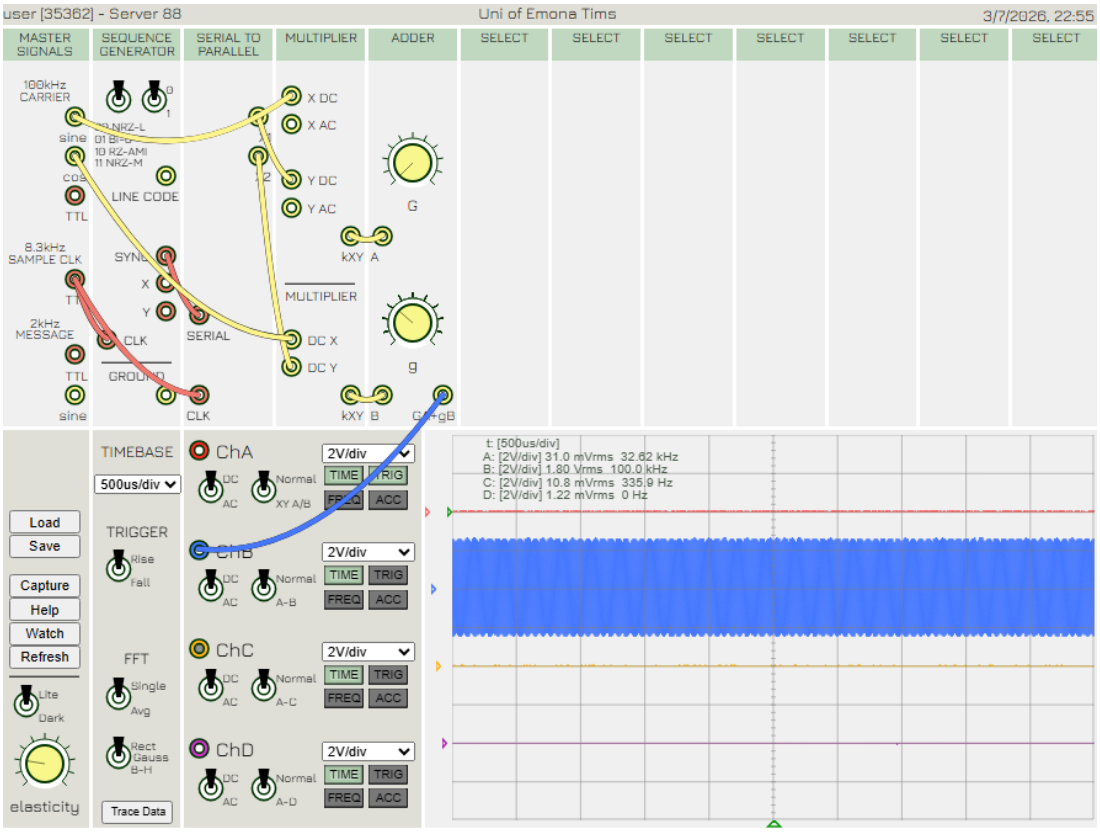
18-A.1


user [35362] - Server 88
Uni of Emona Tims
3/7/2026, 22:51
MASTER SIGNALS
SEQUENCE GENERATOR
SERIAL TO PARALLEL
MULTIPLIER
SELECT
100kHz CARRIER
sine
cos
TTL
8.3kHz SAMPLE CLK
2kHz MESSAGE
TTL
sine
LINE CODE
SYNC
X
Y
CLK
GROUND
CLK
SERIAL
X DC
X AC
Y DC
Y AC
kXY
MULTIPLIER
DC X
DC Y
XY
TIMEBASE
200us/div
TRIGGER
Rise
Fall
FFT
Single
Avg
Rect
Gauss
B-H
Trace Data
Load
Save
Capture
Help
Watch
Refresh
Lite
Dark
elasticity
ChA
ChB
ChC
ChD
2V/div
DC
AC
Normal
XY A/B
A-B
A-C
A-D
TIME
TRIG
FREQ
ACC
t: [200us/div]
A: [2V/div] 53.5 mVrms 181.5 kHz
B: [2V/div] 1.86 Vrms 0 Hz
C: [2V/div] 2.89 Vrms 100.0 kHz
D: [2V/div] 0.00 mVrms 0 Hz

18-A.2

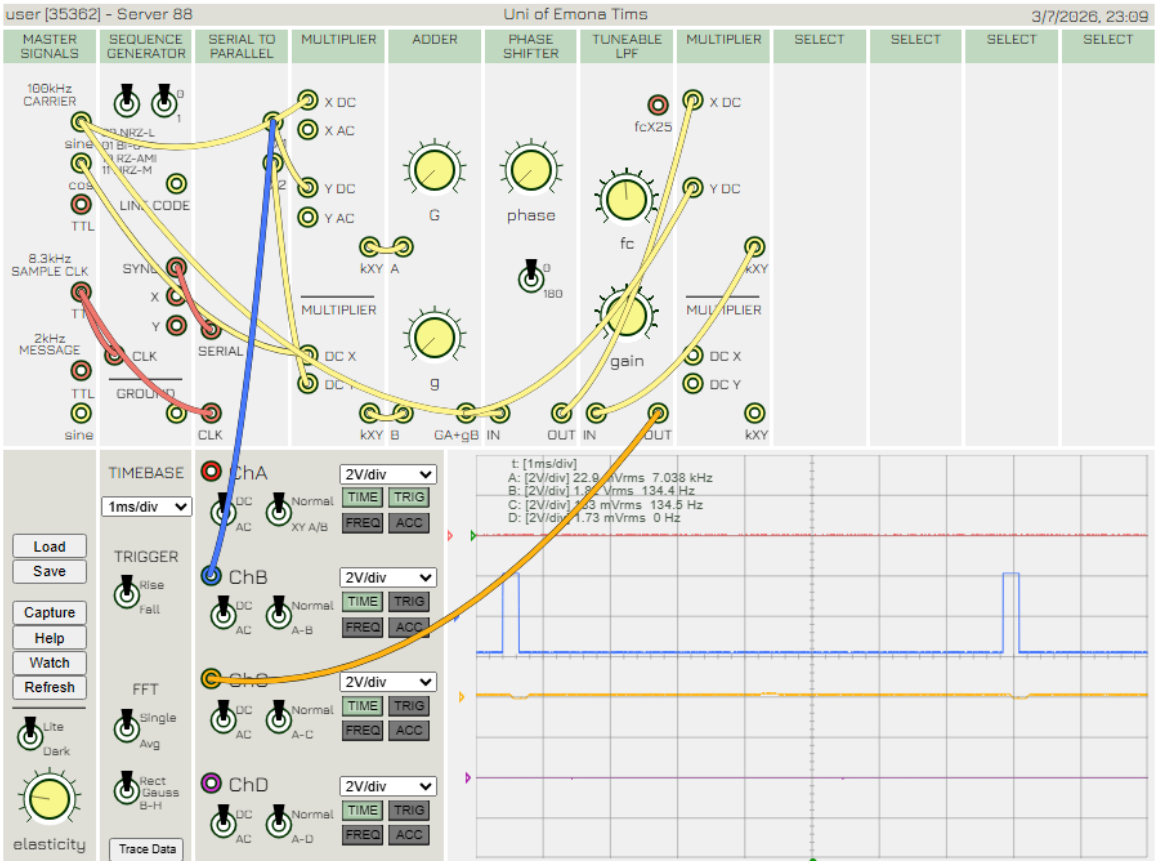

user [35362] - Server 88
Uni of Emona Tims
3/7/2026, 22:52
MASTER SIGNALS
SEQUENCE GENERATOR
SERIAL TO PARALLEL
MULTIPLIER
SELECT
100kHz CARRIER
sine
cos
TTL
8.3kHz SAMPLE CLK
2kHz MESSAGE
TTL
sine
LINE CODE
SYNC
X
Y
CLK
GROUND
CLK
SERIAL
X DC
X AC
Y DC
Y AC
kXY
MULTIPLIER
DC X
DC Y
XY
TIMEBASE
200us/div
TRIGGER
Rise
Fall
FFT
Single
Avg
Rect
Gauss
B-H
Trace Data
Load
Save
Capture
Help
Watch
Refresh
Lite
Dark
elasticity
ChA
ChB
ChC
ChD
2V/div
DC
AC
Normal
XY A/B
A-B
A-C
A-D
TIME
TRIG
FREQ
ACC
t: [200us/div]
A: [2V/div] 40.7 mVrms 174.9 kHz
B: [2V/div] 1.86 Vrms 0 Hz
C: [2V/div] 2.93 Vrms 100.5 kHz
D: [2V/div] 0.00 mVrms 0 Hz

18-A.3

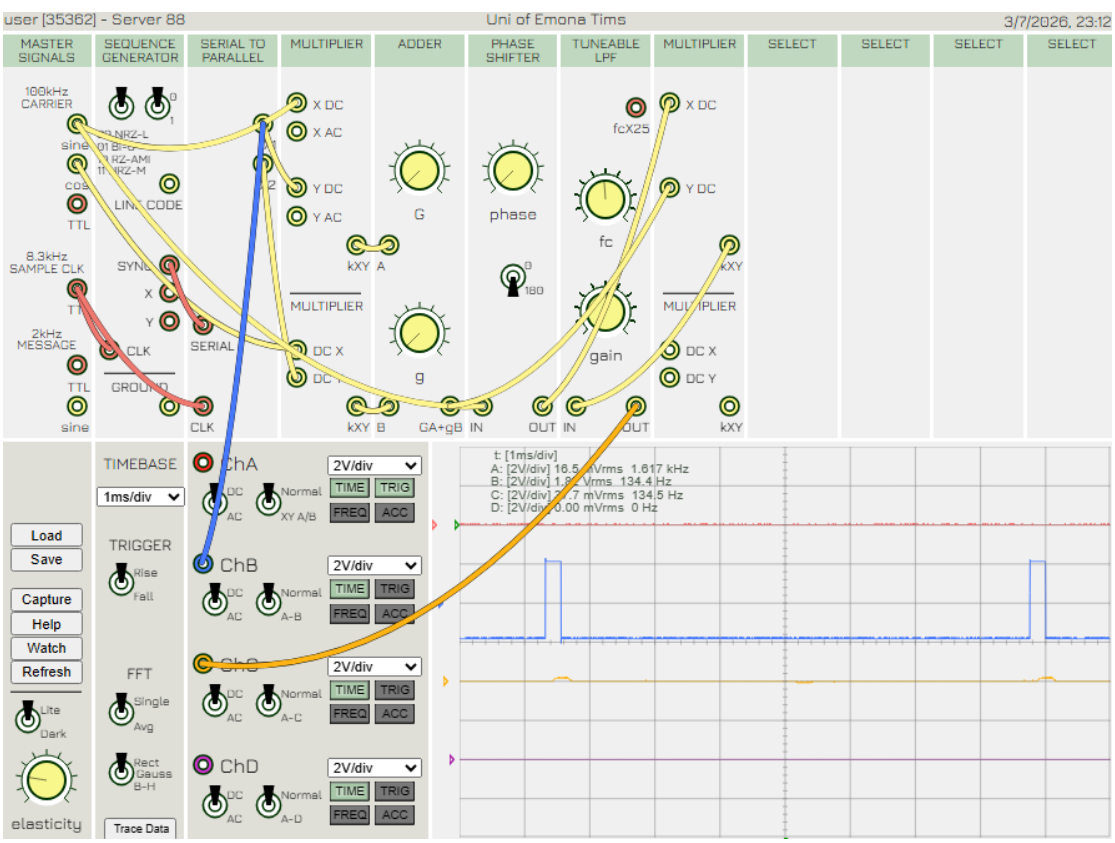

user [35362] - Server 88
Uni of Emona Tims
3/7/2026, 22:55
MASTER SIGNALS
SEQUENCE GENERATOR
SERIAL TO PARALLEL
MULTIPLIER
ADDER
SELECT
100kHz CARRIER
sine
cos
TTL
8.3kHz SAMPLE CLK
2kHz MESSAGE
TTL
sine
LINE CODE
SYNC
X
Y
CLK
GROUND
SERIAL
CLK
X DC
X AC
Y DC
Y AC
kXY A
MULTIPLIER
DC X
DC Y
kXY B
G
g
G+gB
TIMEBASE
500us/div
TRIGGER
Rise
Fall
FFT
Single
Avg
Rect
Gauss
B-H
Trace Data
Load
Save
Capture
Help
Watch
Refresh
Lite
Dark
elasticity
ChA
ChB
ChC
ChD
2V/div
DC
AC
Normal
XY A/B
A-B
A-C
A-D
TIME
TRIG
FREQ
ACC
t: [500us/div]
A: [2V/div] 31.0 mVrms 32.62 kHz
B: [2V/div] 1.80 Vrms 100.0 kHz
C: [2V/div] 10.8 mVrms 335.9 Hz
D: [2V/div] 1.22 mVrms 0 Hz

18-A.4

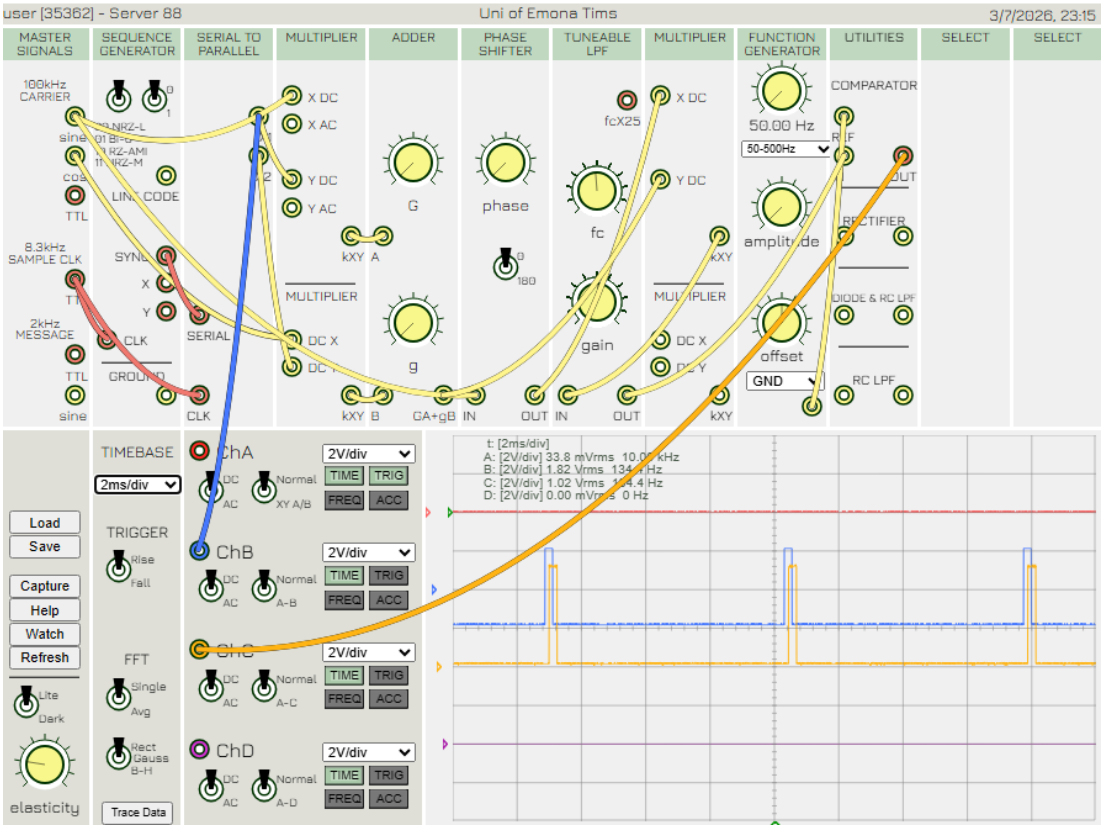
Experiment 18-B (Using phase discrimination to pick-out one of the QPSK signal's BPSK signals)

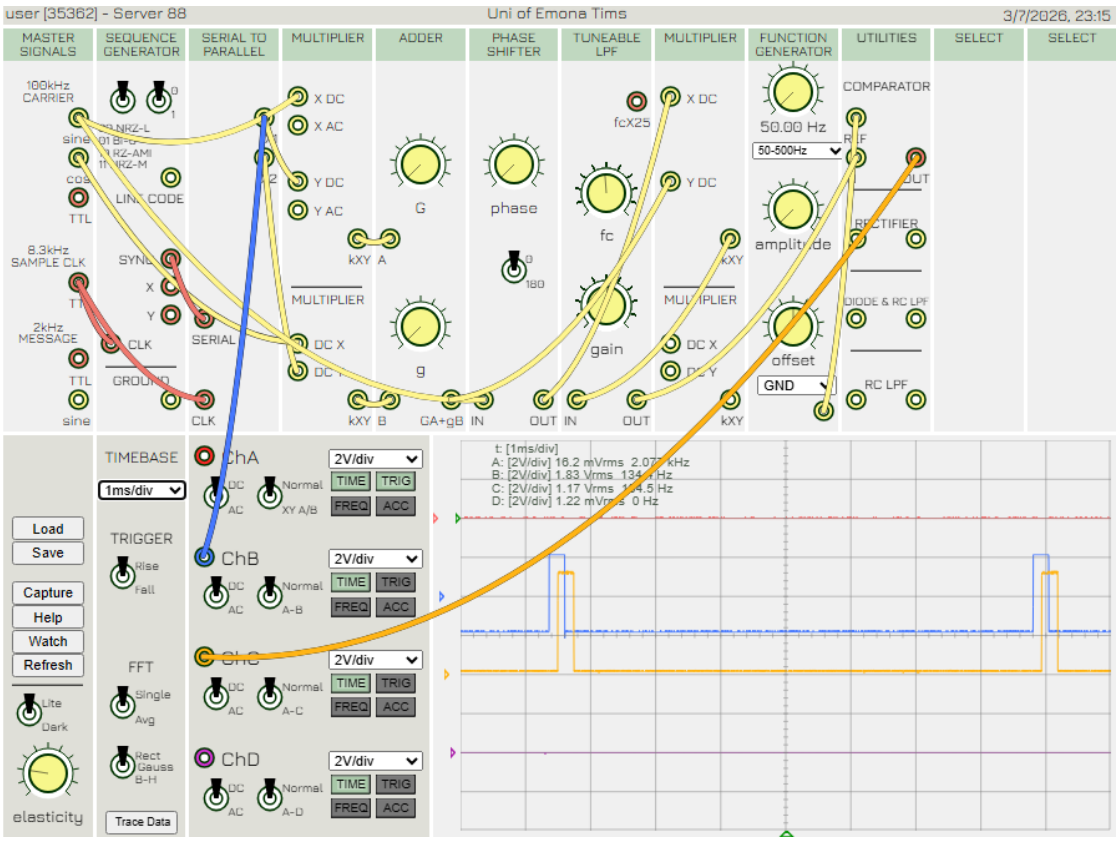


18-B.1 (0 degree phase shift)

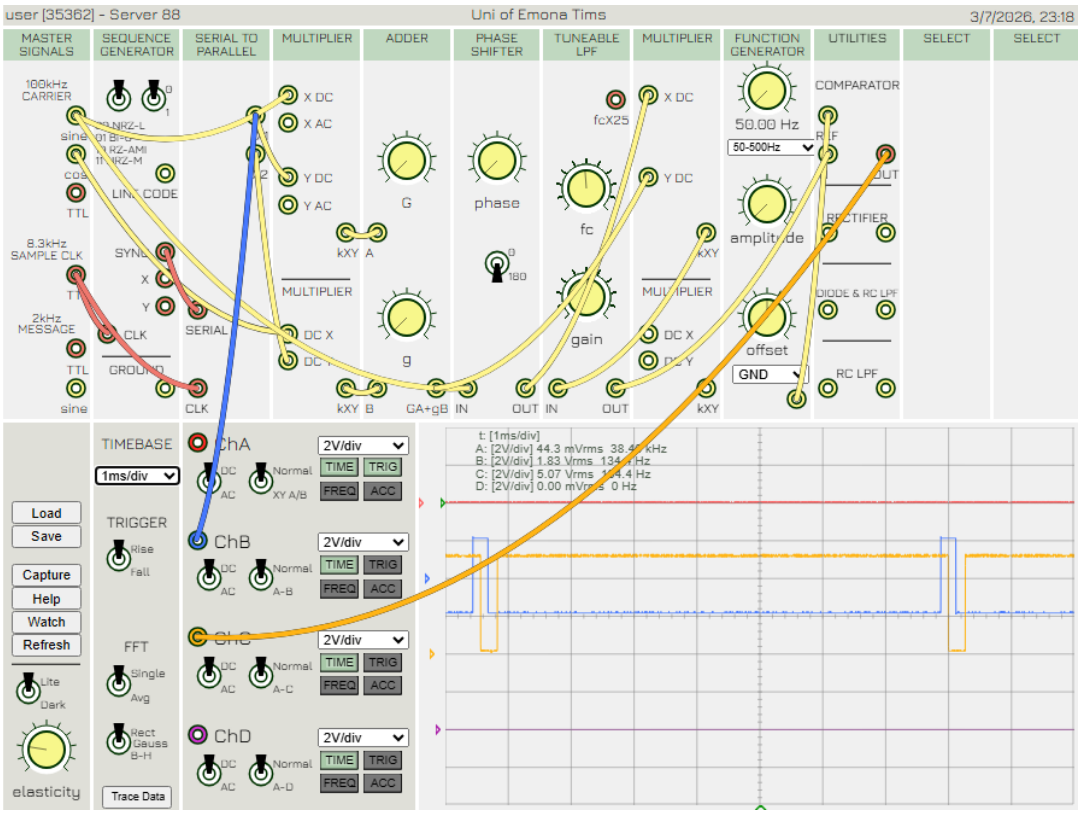


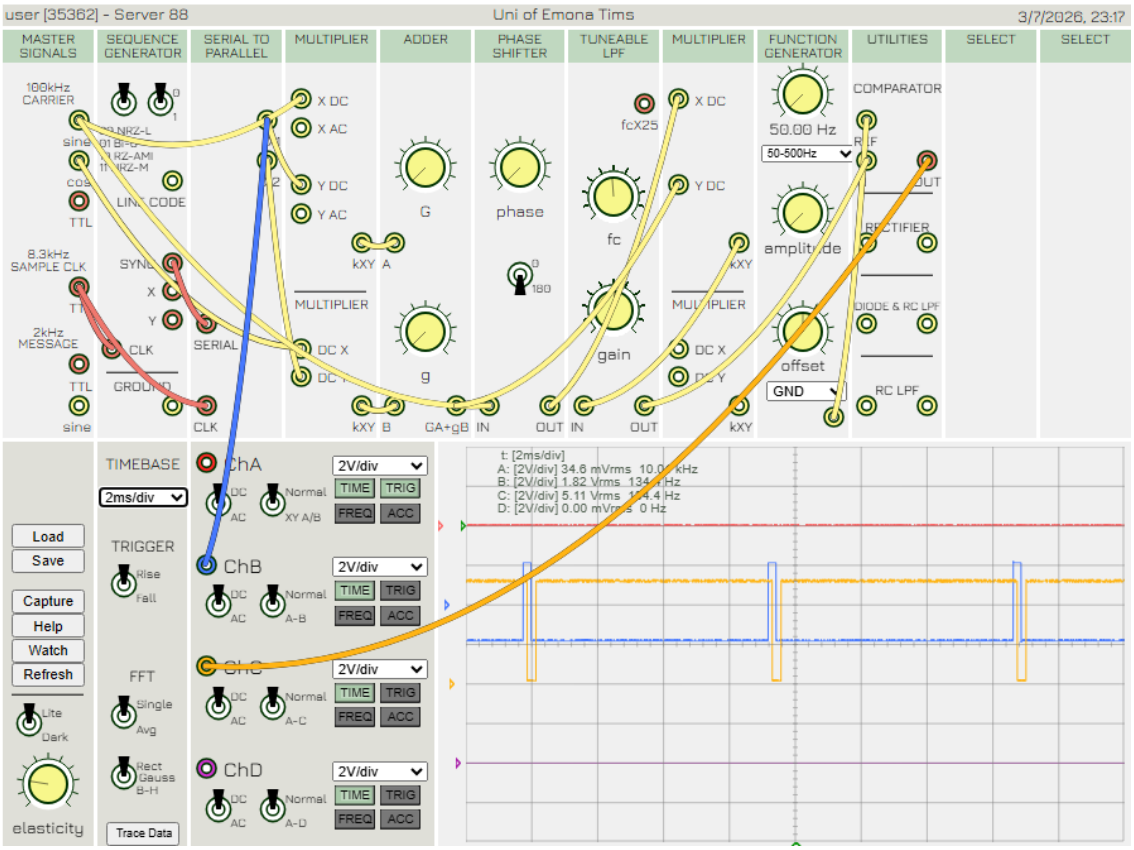
18-B.1 (180 degree phase shift)





18-B.2 (O Degree phase shift)





18-B.2 (180 Degree phase shift)